MILITARY SPECIFICATION

PROJECTILE, 155MM, HE: M483A1 LOADING, ASSEMBLING AND PACKING

This Specification is approved for use within the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the loading, assembling and packing for one type of projectile designated as Projectile, 155MM, HE: M483A1.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

MIL-P-48099 - Propellant, M10 (For use in Special Applications)

STANDARDS

FEDERAL

FED STD-101 - Preservation, Packaging and Packing Materials; Test Procedures

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research, Development and Engineering Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1320
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-286 - Propellants, Solid; Sampling, Examination and Testing

MIL-STD-410 - Nondestructive Testing Personnel Qualification and Certification

MIL-STD-453 - Inspection, Radiographic

MIL-STD-1235- Single and Multi-level Continuous Sampling Procedures and Tables for Inspection by Attributes

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.9)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKING DRAWINGS

9215220 - Projectile, 155MM, HE: M483A1
8837839 - Pallet for Projectile, 155MM, HE: and GB, Hollow Base Type

INSPECTION EQUIPMENT DRAWINGS

8841732 - Test Fixture
9202990 - Operating Instructions

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

CODE OF FEDERAL REGULATIONS

TITLE 49 - Transportation, Parts 100-199

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 100-199 (latest revision)").

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 18 - Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103)

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Projectile. The projectile assembly and all components thereof shall comply with all requirements specified on Drawing 9215220, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Strength of heat seal. The charge bag shall withstand application of a static load, at the seam, of 2.5 pounds per lineal inch, minimum without peel-back of the heat seal and without tensile failure of the area adjacent to the heat seal.

3.4 Ballistic functioning.

3.4.1 Projectile. The projectile shall function satisfactorily (See 6.6).

3.4.2 Grenade. The grenade shall function satisfactorily (See 6.7).

3.4.3 Safety. There shall be no premature burst (See 6.6.1) or evidence thereof, and there shall be no separation of metal parts either in the bore of the weapon or in flight (prior to initiation of the time fuze) in any of the ballistic functioning tests. This requirement shall also apply to those rounds to be fired in the safety phase.

3.5 Radiographic examination of projectiles. Prior to forwarding the projectiles to the proving ground for ballistic testing, the projectiles shall be subjected to radiographic

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

examination. The projectiles shall be examined for the following:

a. Gap between adapter and base (see note).
b. Gap between adapter and grenade (see note).
c. Excessive gap at take-up pad area between grenade and forward plate (see note).
d. Gap between grenades (see note).
e. Safety pin or safety clip shall not be present.
f. Any missing or improperly positioned component or assembly normally discernible on the radiograph.

Note: Gaps in excess of 0.015 inches shall be considered rejects for this inspection.

3.6 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Records of results of contractor-conducted examinations and tests. The contractor shall maintain records of the results of all examinations or tests required by this specification. (See 6.5).

3.8 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease rust, and other foreign matter. In addition, plastic component parts shall be free from splits, cold flow, shrinkage, flash, gouges, cold shuts, inclusion, porosity, or any similar characteristic. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material are permitted. All required marking and stamping shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part description	Drawings	Quantity
Projectile	9215220	12
Steel Shim	9215397	25
Expulsion Charge Assembly	9272019	50
Lid Container, Expulsion Charge	9362552	See Note
Cup Container, Expulsion Charge	9362553	See Note
Container Assembly, Expulsion Charge	9362555	50
Expulsion Charge Assembly	9362556	25

Note: Five (5) samples from each mold cavity.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I herein.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:57Z
Check the source to verify that this is the current version before use.

TABLE I - First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
	Projectile and Components		1 OF 1		See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Projectile (Dwg 9215220)				
	Examination for Defects	12	-	3.2	4.4.2.14
	Ballistic functioning	12		3.4	4.4.3.1
	Expulsion Charge Assembly (Dwg 9272019)				
	Examination for Defects	50		3.2	4.4.2.4
	Steel Shim (Dwg 9215397)				
	Examination for Defects	25		3.2	4.4.2.7
	Lid Container, Expulsion Charge (Dwg. 9362552)	Note			
	Examination for defects	(a)		3.2	4.4.2.16
	Cup Container, Expulsion Charge (Dwg 9362553)	Note (a)			
	Examination for Defects			3.2	4.4.2.17
	Container Assembly Expulsion Charge (Dwg. 9362555)				
	Examination for Defects	50		3.2	4.4.2.19
	Expulsion Charge Assembly (Dwg 9362556)				
	Examination for Defects	25		3.2	4.4.2.20

NOTES: (a) Five (5) samples from each mold cavity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

4.3.3 Rejection. See MIL-A-48078. In addition, the first article sample shall be rejected if the ballistic test results reveal any one of the following:

a. Any premature burst occurs (See 6.6.1).
b. Any separation of metal parts occurs prior to initiation of the time fuze.
c. Any detonation of the projectile cargo occurs prior to initiation of the time fuze.
d. The total number of grenades failing to function satisfactorily (See 6.7.1) equals or exceeds thirty-six (36) (Not applicable to the safety phase).
e. Any projectile fails to function satisfactorily (See 6.6.2 and 6.6.3).

Note 1: Letter d refers to the cumulative number of grenades failing to function satisfactorily in the entire eight (8) rounds tested in the low and high zone phases.

Note 2: A round-by-round accounting shall be made for lost grenades and the replacement provisions of 4.4.3.1.1 utilized when applicable.

Note 3: Grenade air detonations (See 6.7.2) shall be counted and recorded for informational purposes. However, if the total number of air detonations from the entire eight rounds tested in the low and high zone phases exceeds sixteen (16), the data shall be referred to the engineering agency (See 6.4) for decision on final acceptance of the first article.

Note 4: Grenades evaluated to have malfunctioned due to damage resulting from air collision/detonation shall not be included in the total number of grenades used to accept or reject the lot. Grenades so malfunctioning shall be counted and recorded on the evaluation sheet for informational purposes. However, if the total number of such grenades from the entire eight (8) rounds tested in the low and high zone phases exceeds twenty (20), the data shall be referred to the engineering agency (See 6.4) for decision on final acceptance of the first article.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot shall contain:

a. Grenades (M42) from one finally accepted lot interfix number from one manufacturer (See Note 1).

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:57Z
Check the source to verify that this is the currentversion before use.

b. Grenades (M46) from one finally accepted lot interfix number from one manufacturer (See Note 1).
c. Projectile metal parts (body assembly) from one lot interfix number from one manufacturer.
d. Base and cup assemblies from one lot interfix number from one manufacturer.
e. Adapters from one lot interfix number from one manufacturer.
f. Forward plates from one lot interfix number from one manufacturer.
g. Sleeves from one lot interfix number from one manufacturer.
h. Spacers from one lot interfix number from one manufacturer.
i. Splines from one lot interfix number from one manufacturer.
j. Charge assembly propellant from one manufacturer and no more than two serial lot numbers.
k. Expulsion charge assemblies from one lot interfix number from one manufacturer.
l. Loaded projectiles of one weight zone (See Note 2).
m. Lids, expulsion charge container, from one lot interfix number from one manufacturer.
n. Cup, expulsion charge container, from one lot interfix number from one manufacturer.

Note 1: Only those grenade lots which have satisfactorily passed the grenade functioning test shall be assembled into the projectile.

Note 2: Projectiles falling out of the inspection lot criteria due to the zone weight restriction may be accumulated into a hybrid lot provided they originally met all other lotting restrictions during manufacture and the original lot passed all acceptance tests. Further ballistic testing of this hybrid lot is not required.

4.4.2 Examination. See MIL-A-48078.

a Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Continuous sampling in accordance with MIL-STD-1235 may be utilized if approved by the Procuring Activity.

b. Automatic gage. Facilities equipped with Government furnished automatic gaging capable of performing 100% inspection of any characteristic shall utilize this capability in accordance with the contract.

c. Alternate inspection procedures. The contractor may propose the use of alternate inspection procedures in lieu of conducting sampling inspection for defects listed in this specification. (See 6.11).

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:57Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48749D (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9272021
4.4.2.1	Bag, Charge				NEXT HIGHER ASSEMBLY 9272020
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length of bag		0.40%	3.2	Gage
102	Distance to end heat seal max.		0.40%	3.2	Gage
103	Distance to side heat seal		0.40%	3.2	Gage
104	Minimum width of both heat seals over entire length		0.40%	3.2	Gage
105	Heat seal incomplete (2 places)		0.40%	3.2	Visual
106	Strength of heat seal	(a)		3.3	4.5.3
107	Location of eyelet		0.40%	3.2	Gage
108	Location of end stitching		0.40%	3.2	Gage
109	Distance to side stitching		0.40%	3.2	Gage
Minor					
201	Stitching incomplete (2 places)		0.65%	3.2	Visual
202	Fold width eyelet end		0.65%	3.2	Gage
203	Fold misalignment max		0.65%	3.2	Gage
204	Thickness of assembled eyelet		0.65%	3.2	Gage
205	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

(a) Sample 20 per lot, reject on one (1) or more defectives.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48749D (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Cover, Expulsion Charge	1 OF 1	9272022
			NEXT HIGHER ASSEMBLY 9272020

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of rim		0.40%	3.2	Gage
102	Thickness of rim, minimum		0.40%	3.2	Gage
103	Total height		0.40%	3.2	Gage
104	Angle between rim and angled portion		0.40%	3.2	Gage
105	Thickness of angled portion, minimum		0.40%	3.2	Gage
106	Height of rim		0.40%	3.2	Gage
107	Location of tabs		0.40%	3.2	Gage
108	Distance between tab and angled portion (4 places)		0.40%	3.2	Gage
109	Radius missing		0.40%	3.2	Visual
110	Crack or puncture		0.40%	3.2	Visual
111	Burr		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48749D (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9272020
4.4.2.3	Charge Loading Assembly				NEXT HIGHER ASSEMBLY 9272019
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pull wire or bag loose with respect to cover		0.40%	3.2	Visual/Manual
102	Cover damaged		0.40%	3.2	Visual
103	Clinch of stud improper		0.40%	3.2	Visual
104	Bag damaged		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9272019
4.4.2.4	Expulsion Charge Assembly				NEXT HIGHER ASSEMBLY 9215220
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Leak test	(a)		3.2	4.5.2
102	Weight of propellant charge incorrect		0.40%	3.2	Gage
103	Bag seal incomplete (3 places)		0.40%	3.2	Visual
104	Moisture and total volatiles content of propellant	(a)		3.2	4.5.1
105	Bag damaged		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

(a) Sample in accordance with the applicable test method paragraph.

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T03:57Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9215380
4.4.2.5	Adapter				NEXT HIGHER ASSEMBLY 9215220
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Verification of heat treatment of adapter		100%	3.2	4.5.4
Major	None defined				
Minor	None defined				

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 10542902-1
4.4.2.6	Projectile Metal Parts Assembly, Prior to Loading		NEXT HIGHER ASSEMBLY 9215220

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Threads damaged at base end		0.40%	3.2	Visual
102	Threads damaged at nose end		0.40%	3.2	Visual
103	Internal lubricant applied improperly		0.40%	3.2	Visual
Minor					
201	Foreign matter on interior surface of body		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48749D (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9215397
4.4.2.7	Shim, Steel				NEXT HIGHER ASSEMBLY 9215220
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Hardness		0.40%	3.2	4.5.5
102	Diameter, min.		0.40%	3.2	Gage
103	Thickness, min.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 — Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil - Downloaded: 2026-02-26T03:57Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48749D (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Assembly, Forward Plate and O-Ring				None
					NEXT HIGHER ASSEMBLY 9215220
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	O-ring not greased		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.9	Grenade Assembly, Prior to Loading Into Projectile				9215340 or 9215370 as applicable
					NEXT HIGHER ASSEMBLY 9215220
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Safety pin missing		100%	3.2	Visual
2	Slider lock cracked, incomplete, missing or improperly positioned		100%	3.2	Visual
Major					
101	Safety clip not removed		0.40%	3.2	Visual
102	Tape unwound		0.40%	3.2	Visual
103	Tape not completely inserted in slots of stiffener		0.40%	3.2	Visual
Minor	None defined				

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48749D (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	Projectile, After Removal of Safety Pins From Grenade Assembly, After Each Layer is Inserted Into the Projectile				None
					NEXT HIGHER ASSEMBLY 9215220
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Slide assembly protruding from grenade fuze housing indicating armed or partially armed (all layers)		100%	3.2	Visual
2	Grenade type incorrect as evidenced by protective coating or lack of yellow marking (9-11 layers only)		100%	3.2	Visual
3	Slider lock cracked, incomplete, missing or improperly positioned		100%	3.2	Visual
4	Grenade missing or improperly positioned (all layers)		100%	3.2	Visual
5	Any sleeve, spacer or spline missing, improperly positioned or incorrect length or wrong color (all layers)		100%	3.2	Visual
Major					
101	Grenade type incorrect (1-8 layers only)		0.40%	3.2	Visual
102	Safety pin not removed from each grenade (all layers)		0.40%	3.2	Visual
103	Stiffener or tape improperly positioned (all layers)		0.40%	3.2	Visual
Minor	None defined				

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48749D (AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.11	Projectile and Base, After Assembly of Adapter and Immediately Prior to Assembly of Base and Seal Ring		1 OF 1	None
				NEXT HIGHER ASSEMBLY 9215220

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Adapter missing or improperly assembled		100%	3.2	Visual
2	Any sleeve, spacer or spline missing, improperly positioned or incorrect length or wrong color		100%	3.2	Visual
Major					
101	Depth, between projectile face and adapters or shims as applicable at specified load		100%	3.2	4.5.11
102	Seal ring missing or wrong type		0.40%	3.2	Visual
103	Seal ring not greased		0.40%	3.2	Visual
104	Sealant missing on threads of base		0.40%	3.2	Visual
105	Threads damaged on base of projectile body		0.40%	3.2	Visual
106	Foreign material on threads		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 — Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Projectile, Prior to Assembly of Expulsion Charge Assembly, and Grommet Assembly				None
					NEXT HIGHER ASSEMBLY 9215220
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
a.	Gap between projectile body and base 360°, max.		100%	3.2	Gage
b.	Bare spots in anodic hard coat on base.		100%	3.2	Visual
Major					
101	Disassembly torque test of base	(a)	0.40%	3.2	4.5.8
102	Depth from surface of projectile to expulsion charge cup, min.		0.40%	3.2	Gage
103	True position of min. internal diameter of expulsion charge cup to pitch diameter of ogive threads		0.40%	3.2	Gage
104	Expulsion charge cup improperly staked		0.40%	3.2	Visual/Manual
105	Obturator missing or damaged		100%	3.2	Visual
106	Evidence of axial movement or obturator		100%	3.2	Manual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES: (a) Sample size per MIL-STD-105, Inspection Level I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48749D (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.13	Projectile, Prior to Assembly of Grommet Assembly Gasket and Lifting Plug Assembly				None
					NEXT HIGHER ASSEMBLY
					9215220
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special a.	Weight		100%	3.2	Gage
Major 101	Expulsion charge missing		0.40%	3.2	Visual
Minor 201	Evidence of poor workmanship		0.65%	3.8	Visual
202	Unpainted surfaces of lifting plug not coated with anti-seize compound (DWG 9345325)		0.65%	3.2	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T03:57Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48749D (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9215220
4.4.2.14	Projectile				NEXT HIGHER ASSEMBLY None
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Lifting plug assembly wrong type (not fusible)		100%	3.2	Visual
Special					
a.	Lifting plug or gasket missing or damaged		100%	3.2	Visual
b.	Bare spots on protective coating		100%	3.2	Visual
Major					
101	Disassembly torque tests of lifting plug assembly	(a)	0.40%	3.2	4.5.9
102	Air pressure test	(b)	0.40%	3.2	4.5.10
103	Diameter of bourrelet, max (applicable after paint on projectile has been retouched)		100%	3.2	Gage
104	Radiographic examination	(c)		3.5	4.5.9
105	Zone marking missing, incorrect or unidentifiable		0.40%	3.2	Visual
106	Diamond marking missing, incorrect or unidentifiable		0.40%	3.2	Visual
Minor					
201	Any part missing or damaged		0.65%	3.2	Visual
202	Rotating band damaged		0.65%	3.2	Visual

NOTES:
(a) Sample size per MIL-STD-105, Inspection Level I.
(b) Sample size per MIL-STD-1235, CSP-1, Code Letter F.
(c) Ballistic samples only.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.14	Projectile				9215220
					NEXT HIGHER ASSEMBLY: None
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor (cont)					
203	Marking missing or unidentifiable		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.15	Pallet Assembly				8837839
					NEXT HIGHER ASSEMBLY: None
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined	None	defined		
Major					
101	Incorrect strapping or improperly assembled		0.40%	3.2	Visual
102	Seal strapping missing		0.40%	3.2	Visual
103	Lifting plug of projectile missing or damaged		0.40%	3.2	Visual
Minor					
201	Board broken or split		0.65%	3.2	Visual
202	FSN number missing or unidentifiable		0.65%	3.2	Visual
203	Any other marking missing incorrect or unidentifiable		0.65%	3.2	Visual

NOTES:

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.16	Lid, Container, Expulsion Charge				9362552
					NEXT HIGHER ASSEMBLY 9362556
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Inside diameter of locking lip		0.40%	3.2	Gage
103	Inside diameter of open end		0.40%	3.2	Gage
104	Inside diameter of closed end		0.40%	3.2	Gage
105	Length of cylindrical portion of lid		0.40%	3.2	Gage
106	Incomplete or missing tab		0.40%	3.2	Visual
107	Profile of locking lip area (destructive)	(a)		3.2	Gage
108	Total height of lid		0.40%	3.2	Gage
109	Depth of cap		0.40%	3.2	Gage
110	Thickness of tab below ridges		0.40%	3.2	Gage
111	Width of ribs		0.40%	3.2	Gage
112	Height of ribs		0.40%	3.2	Gage
113	Pull test, min. and max.	(b)		3.2	4.5.12
Minor					
201	Missing slot (4 places)		0.65%	3.2	Visual
202	Presence of friction ribs on tab		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:
(a) Sample three (3) parts from each mold cavity at the end of each shift. No failures allowed.
(b) Sample ten (10) per lot: Reject on one or more failures.

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T03:57Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.17	Cup, Container, Expulsion Charge				9362553
					NEXT HIGHER ASSEMBLY 9362555
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length of cup		0.40%	3.2	Gage
102	Length of cap interlock area		0.40%	3.2	Gage
103	Outside diameter of ribs		0.40%	3.2	Gage
104	Outside diameter of locking lips		0.40%	3.2	Gage
105	Inside diameter at top of cup		0.40%	3.2	Gage
106	Outside diameter at bottom of cup between ribs		0.40%	3.2	Gage
107	Thickness of bottom		0.40%	3.2	Gage
108	Profile of locking lip area		0.40%	3.2	Gage
109	Pull test, min. and max.	(a)		3.2	4.5.12
110	Flatness on seal area		0.40%	3.2	Gage
111	Wall thickness variation between ribs from top to bottom of cup		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

(a) Sample ten (10) per lot: Reject on one or more failures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.18	Disc, Expulsion Charge				9362554
					NEXT HIGHER ASSEMBLY
					9362555
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness		0.40% (a)	3.2	Gage
102	Diameter		0.40% (a)	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES: (a) .40% AQL or alternatively, if the disc is used to seal the expulsion cup immediately after the disc is made, then take 5 samples per machine at the start of production each day and 5 samples per machine after every 4 hours of production or portion thereof If any part fails to comply with the requirement, the parts represented by the sample since the last satisfactory inspection shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.19	Container Assembly, Expulsion Charge		9362555
			NEXT HIGHER ASSEMBLY 9362556

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Leak Test	(a)		3.2	4.5.2.1
102	Weight of propellant charge incorrect		0.40%	3.2	Gage
103	Disc seal incomplete		0.40%	3.2	Visual
104	Moisture and total volatiles content of propellant	(a)		3.2	4.5.1
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

(a) Sample in accordance with applicable test method paragraph.

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48749D (AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.20	Expulsion Charge Assembly				9362556
					NEXT HIGHER ASSEMBLY 9215220
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cup or lid damaged		0.40%	3.2	Visual
102	Gap between cup and lid 360°, max		0.40%	3.2	Gage
103	Total length, max.		0.40%	3.2	Gage
104	Outside diameter of container lid, max		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check for source to verify that this is the current version before use.

4.4.3 Testing. Testing is required as described herein and in the First Article and Quality Conformance Inspection Table.

4.4.3.1 Ballistic functioning (see 3.4) major defect. For first article and for first lot produced and continuing until three (3) consecutive acceptable lots have been accepted, the Government inspector shall select twelve (12) projectiles from each lot for this test. They shall be tested per 4.5.10 as follows:

a. Four (4) projectiles tested at the low zone phase.
b. Four (4) projectiles tested at the high zone phase.
c. Three (3) projectiles tested at the safety phase.

4.4.3.1.1 Projectile replacement. The remaining projectile shall be fired for replacement or retest if any of the following occurs:

a. Low bursting rounds. If any projectile, except those tested at the safety phase, functions below six hundred (600) feet, and the total number of all grenade defects for the 2 zones does not exceed the specified acceptance criteria, the projectile shall be evaluated to function satisfactorily and no replacement for that projectile will be required. However, when the results of the low-bursting projectile would cause the lot represented by the sample to be rejected, the results of that projectile shall be recorded and a replacement projectile shall be fired in the same charge zone. The results of the replacement projectile shall be substituted for those of the low-bursting round in the determination of lot acceptance.

b. Lost grenades. A replacement projectile is required whenever more than ten (10) grenades from one projectile (except those fired at the safety phase) are considered to be lost. Results of these rounds with excesssive lost items shall be recorded for informational purposes. The replacement shall be fired at the same charge zone and the results of the replacement substituted for use in the determination of lot acceptance.

c. Time fuze malfunctions. Time fuze malfunctions shall not be considered as cause for rejection of a projectile lot. When a time fuze malfunction occurs, the proving ground may fire a replacement projectile in the same charge zone.

4.4.3.1.2 Retest. If, for any reason, the proving ground considers that test conditions have affected all or part of the test results, the proving ground shall declare the affected results invalid and perform new tests on additional projectiles, as required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

4.4.3.1.3 Use of twelfth projectile. If the projectile is not required for replacement or retest, it shall be tested at the safety phase and the test results recorded.

4.4.3.1.4 Rejection. The lot represented by the sample shall be rejected if the ballistic test results for that lot reveal any one of the following defects.

a. Any premature burst occurs (See 6.6.1).
b. Any separation of metal parts occurs prior to initiation of the time fuze.
c. Any detonation of the projectile cargo occurs prior to initiation of the time fuze.
d. The total number of grenades failing to function satisfactorily (See 6.7.1) equals or exceeds thirty-six (36). (Not applicable to the safety phase).
e. Any projectile fails to function satisfactorily (See 6.6.2 and 6.6.3).

Note 1: Letter d refers to the cumulative number of grenades failing to function satisfactorily in the entire eight (8) rounds tested in the low and high zone phases.

Note 2: A round by round accounting shall be made for lost grenades and the replacement provisions utilized when applicable.

Note 3: Grenade air detonations (See 6.7.2) shall be counted and recorded for informational purposes. However, if the total number of air detonations from the entire eight (8) rounds tested in the low and high zone phases exceeds sixteen (16) the data shall be referred to the engineering agency (See 6.4) for decision on final acceptance of the lot.

NOte 4: Grenades evaluated to have malfunctioned due to damage resulting from air collision/detonation shall not be included in the total number of grenades used to accept or reject the lot. Grenades so malfunctioning shall be counted and recorded on the evaluation sheet for informational purposes. However, if the total number of such grenades from the entire eight (8) rounds, tested in the low and high zone phases exceeds twenty (20), the data shall be referred to the engineering agency (See 6.4) for decision on final acceptance of the lot.

4.4.3.2 Ballistic functioning (See 3.4) major defect, reduced testing. After three (3) consecutive acceptance lots have met the criteria of 4.4.3.1, the Government inspector shall select six (6) projectiles from each lot for this test. They shall be tested per 4.5.10 as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

a. Two (2) of the projectiles shall be tested at the low zone phase.
b. Two (2) of the projectiles shall be tested at the high zone phase.
c. One (1) of the projectiles shall be tested at the safety phase.

4.4.3.2.1 Projectile replacement. The remaining projectile shall be fired for replacement or retest if any of the following occurs:

a. Low bursting rounds. If any projectile, except those tested at the safety phase, functions below six hundred (600) feet, and the total number of grenade defects for the 2 zones does not exceed the specified acceptance criteria, the projectile shall be evaluated to function satisfactorily and no replacement for that projectile will be required. However, when the results of the low bursting projectile cause the lot represented by the sample to be rejected, the results of that projectile shall be recorded and a replacement projectile shall be fired in the same charge zone. The results of the replacement projectile shall be substituted for those of the low bursting round in the determination of lot acceptance.

b. Lost grenades. A replacement projectile is required whenever more than ten (10) grenades from one projectile (except those fired at the safety phase) are considered to be lost. Results of the rounds with excessive lost items shall be recorded for informational purposes. The replacement shall be fired at the same charge zone and the results of the replacement substituted for use in the determination of lot acceptance.

c. Time fuze malfunctions. Time fuze malfunctions shall not be considered as cause for rejection of a projectile lot. When a time fuze malfunction occurs, the proving ground may fire a replacement projectile in the same charge zone.

4.4.3.2.2 Retest. If for any reason the proving ground considers that test conditions have affected all or part of the test results, the proving ground shall declare the affected results invalid and perform new tests on additional projectiles, as required.

4.4.3.2.3 Use of sixth projectile. If the projectile is not required for replacement or retest it shall be tested at the safety phase and the test results recorded.

4.4.3.2.4 Rejection. The lot represented by the sample shall be rejected (See Note 5) if the ballistic test results for that lot reveal any of the following defects:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

a. Any premature burst occurs (See 6.6.1).
b. Any separation of metal parts occurs prior to initiation of the time fuze.
c. Any detonation of the projectile cargo occurs prior to initiation of the time fuze.
d. The total number of grenades failing to function satisfactorily (See 6.7.1) equals or exceeds eighteen (18) (not applicable to the safety phase).
e. Any projectile fails to function satisfactorily (See 6.6.2 and 6.6.3).

Note 1: Letter d refers to the cumulative number of grenades failing to function satisfactorily in the entire four (4) rounds tested in the low and high zone phases.

Note 2: A round-by-round accounting shall be made for lost grenades and the replacement provisions utilized when applicable.

Note 3: Grenade air detonations (See 6.7.2) shall be counted and recorded for informational purposes. However, if the total number of air detonations from the entire four (4) rounds tested in the low and high zone phases, exceeds eight (8) the data shall be referred to the engineering agency (See 6.4) for decision on final acceptance of the lot.

Note 4: Grenades evaluated to have malfunctioned due to damage resulting from air collision/detonation shall not be included in the total number of grenades used to accept or reject the lot. Grenades so malfunctioning shall be counted and recorded on the evaluation sheet for informational purposes. However, if the total number of such grenades from the entire four (4) rounds tested in the low and high zone phases exceeds ten (10), the data shall be referred to the engineering agency (See 6.4) for decision on final acceptance of the lot.

Note 5: If any lot is rejected, sampling and acceptance for ballistic functioning shall revert back to the criteria of 4.4.3.1.

4.4.3.3 Dimensional control of plastic parts. In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, and after a curing time for the parts has been approved (see Note 1), a sample of three (3) parts (as molded) from each cavity may be dimensionally inspected to qualify a new or reworked cavity for use in production or as a control of each cavity during production.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

The inspection of the parts shall be performed only when the curing time for the parts has been reached. The inspection for control of each cavity during production shall be performed on the above quantities from each cavity after production of each 500 parts or at the end of each shift, providing the number of parts produced during the shift is less than 500 parts. Of the three (3) samples, one (1) sample shall be the last part produced. Parts that fail to comply with dimensional requirements after dimensional stability has been attained shall be classed defective.

If any defective parts are found during qualification of the cavity, the cavity producing the defective parts will not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for corrective action.

All cavities formerly removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test as above. The contractor may request a change of inspection frequency providing he presents objective evidence to substantiate the request to the Contracting Officer.

The contractor shall furnish evidence to the Government inspector that dimensional stability has been attained prior to measuring the parts for acceptance.

Note (1) - Establishing a curing time as follows. Dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30 or 60 mins.) until dimensional stability is attained. The curing time will be from the time that part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to Commander, US Army AMCCOM, ATTN: AMSMC-QAR-Q(D), Dover, NJ 07801-5001. If any new change in material, cycle time, a reworked cavity, or new cavity is used, a new curing time shall be established and approved.

4.4.4 Inspection equipment. For the inspection equipment required to perform the examinations and tests prescribed herein the contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, 6.3 and 6.10 herein.

4.5 Methods of inspection.

4.5.1 Moisture and total volatiles content of propellant. The moisture and total volatiles content of each lot of propellant

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

or portion thereof shall be determined for each eight (8) hours of production. This determination shall be accomplished prior to the anticipated use of the lot or portion thereof, with a maximum of eighteen (18) hours lead time. Sampling of the propellant shall be in accordance with MIL-STD-286, Method 101.1.2. If the moisture and total volatiles content fails to comply with the requirements of MIL-P-48099, the propellant lot shall be rejected until corrective action has been taken as verified by repeating the moisture and total volatiles content determination. The contractor shall provide adequate controls to insure maintaining compliance with the requirements between testing and use.

4.5.2 Leak test of expulsion charge assembly (dwg. 9272019 (destructive test). A sample of five (5) assemblies from each hours production per sealing machine shall be selected for this test. Each sample assembly shall be subjected to the vacuum specified on the applicable drawing using equipment and procedures shown on drawings 8841732 and 9202990 (except test instruction 4, see Note 1 for reject criteria). If any sample assembly fails the leak test, (see Note 1) the quantity represented by that sample shall be set aside and retested per the following procedure:

a. Visually inspect all bags for completeness of heat seals (top, side, and bottom). Bags with damaged or incomplete seals shall be classed defective and removed from production.

b. From remaining assemblies, ten (10) shall be selected and subjected to the leak test, reject on one or more defects.

Note 1: For the purpose of this test a reject shall be defined as an assembly that leaks to the extent that the M10 propellant becomes wet. To determine the condition of the propellant those assemblies that exhibit bubbles during the test shall be cut open and the propellant examined (visually and manually) for moisture. Only those assemblies containing moisture shall be considered defective.

4.5.2.1 Leak test of expulsion charge container assembly (dwg. 9362555) (destructive test). A sample of five (5) assemblies (without lids) from each hour's production per sealing machine shall be selected for this test. Each sample assembly shall be subjected to the vacuum specified on the applicable drawing using equipment and procedures shown on drawings 8841732 and 9202990 (except test instruction 4, see Note 1 for reject criteria). If any sample assembly fails the leak test (see Note 1), the quality represented by that sample shall be set aside and retested per the following procedure:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

a. Visually inspect all cups for completeness of heat seals. Cups with damaged or incomplete seals shall be classed defective and removed from production.

b. From remaining assemblies, ten (10) shall be selected and subjected to the leak test, reject on one or more defects.

Note 1: For the purpose of this test, a reject shall be defined as an assembly that leaks to the extent that the M10 propellant becomes wet. To determine the condition of the propellant those assemblies that exhibit bubbles during the test shall be cut open and the propellant examined (visually and manually) for moisture. Only those assemblies containing moisture shall be considered defective.

4.5.3 Strength of heat seal. This test shall be performed in accordance with FED-STD-101, Method 2024, as modified herein. One specimen, one inch wide, shall be cut perpendicular to the line of a seam of each sample bag. Location of the sample shall be random so that both the end seam and the long seam are sampled. One leg of the specimen shall be clamped so that the specimen hangs freely. The required load (2.5 lbs) shall be applied to the free end in such a manner as to avoid impact loading the seam. The load shall be allowed to act for a minimum of one minute after which the seam shall be examined for peel - back of the heat seal.

NOTE: The test samples shall be conditioned prior to testing at a temperature of 70 to 80 degrees fahrenheit (F) and shall be tested under these conditions unless otherwise specified.

4.5.4 Verification of heat treatment of adapter (non-destructive). This inspection shall employ the eddy current method, or any other equivalent method approved by the procurement agency. If the eddy current method is used, all personnel operating and calibrating this equipment shall be qualified and certified in accordance with MIL-STD-410. The hardness limits shall be established and submitted to the COR for approval prior to use. If any adapter fails to comply with the requirements specified on the applicable drawing, it shall be classed defective and removed from production.

4.5.5 Hardness test of shim (dwg. 9215397) (non-destructive). This test shall be performed in accordance with ASTM E 18 using the standard penetrator for the specified hardness scale. Three readings shall be taken approximately 120° apart. If any reading fails the requirement on the applicable drawing that shim shall be classed defective.

4.5.6 Disassembly torque test of base. The projectile with base assembled, shall be subjected to the disassembly torque specified on the applicable drawing. Any unit that fails to withstand the disassembly torque shall be classed defective.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

4.5.7 Disassembly torque test of lifting plug assembly. The projectile with lifting plug assembled, shall be subjected to the disassembly torque specified on the applicable drawing. Any unit that fails to withstand the disassembly torque shall be classed defective.

4.5.8 Air pressure test of projectile. The assembled projectile, with lifting plug, shall be placed in one or more chambers, (the base to body, body to ogive and ogive to lifting plug joints must be tested) and a pre-determined volume of air shall be delivered to produce the specified pressure between the inside of the chamber and the outside of the projectile. After initiation of test, the pressure must achieve the required level and maintain that level for the specified time interval. Failure to meet either criteria shall be cause for rejection, that projectile shall be suitably identified and removed from the lot to prevent mixing with accepted projectiles.

All equipment used in this test shall be capable of performing the test accurately and consistently. Equipment shall be calibrated at the start of each shift and thereafter, at intervals of not more than four (4) hours of continuous operation. It shall also be calibrated prior to restarting the operation after a layoff of three (3) hours or more. If it is found that the gage is out of calibration in such a direction that the test pressure was actually lower than that specified on the drawing, projectiles tested since the last satisfactory calibration shall be retested after the gage error has been corrected.

4.5.9 Radiographic inspection of projectile. The projectile assemblies shall be radiographically examined prior to forwarding to the proving ground. Prior to assembly of the base, the Government inspector shall indicate the location of the spline by marking the outside of the projectile. In all exposures the radiographic beam shall be focused along the center axis of the projectile and parallel to the face of the forward plate, with the spline ninety degrees (90°) to the right in the direction from the source to the projectile. Radiographic exposure shall as a minimum, include all the base material adjacent to the adapters, to and including the entire forward plate. If necessary, more than one radiograph shall be taken in this view as required to avoid detrimental parallax effects and to observe the conditions specified in 3.5. In the event the radiograph(s) taken in this view prevent a suitable examination, the projectile shall again be radiographed at a location rotated 90° from the original with the spline nearest the source. Projectiles and radiographs shall be suitably identified to permit cross reference. Radiographs shall be maintained by the contractor until final acceptance of the projectile lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

If any defect, per 3.5 is found, it shall be reported to AMSMC-QAR-Q(D). The lot represented by this sample shall be rejected.

NOTE: Radiographic equipment, operations and procedures shall be qualified in accordance with MIL-STD-453. The personnel performing this examination shall be qualified and certified, level 2, per MIL-STD-410. In addition, the radiographic examination shall be performed with a technique capable of producing 2 percent sensitivity (radiographic quality level 2-2T) and the photographic density of the film used shall be of a quality which will produce a sharp definition of internal structure.

4.5.10 Ballistic functioning tests. These tests shall be conducted at a Government proving ground in accordance with Aberdeen Proving Ground Acceptance Test Procedure LO-VAR and Supplement LO-155H-5J. The round shall be fitted with a M577 fuze. The projectile fuze setting and quadrant elevation of the weapon shall be adjusted to attempt to achieve ejection, at the target height specified below, over a prepared impact area. The maximum ejection height shall be eighteen hundred (1800) feet, for any zone. A photograph shall be taken of the fuze setting for each round immediately prior to use and identified with the round number. The sample projectile shall be tested at the specified zones using weapons, equipment methods and procedures in accordance with the acceptance test procedures referenced above. Tubes used for this test shall not have less than twenty-five (25) percent remaining tube life based on current gun tube condemnation criteria.

4.5.10.1 Low zone. The sample projectiles to be subjected to this test shall be fired in the M185 Cannon, using the Zone 3 propelling charge (M3A1). The target ejection height for this phase shall be nine hundred (900) feet. Observation shall be made for failure to comply with the requirements of 3.4.

4.5.10.2 High zone. The sample projectiles to be subjected to this test shall be fired in the M185 Cannon, using the M119 series propelling charge. The target ejection height for this phase shall be fifteen hundred (1500) feet. Observation shall be made for failure to comply with the requirements of 3.4.

4.5.10.3 Safety phase. The sample projectiles to be subjected to this test shall be fired at ambient temperature in the M199 Cannon after modification of the M203 propelling charge by removing sufficient propellant to yield a chamber pressure of 43 to 46 thousand pounds per square inch (KPSI). Observation shall be made for failure to comply with the requirements of 3.4.3. Burst height and grenade functioning are not considered requirements of this test, but shall be reported for informational purposes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

4.5.10.4 Records of ballistic tests. A record shall be made for each round fired and shall include an accounting for the defects listed in 4.3.3 or 4.4.3.1.4 or 4.4.3.2.4 as applicable. In addition, a record shall be made of fuze duds, fuzes with excessive off-times, and any other malfunctions. A copy of the records of all ballistic test results shall be forwarded to the address specified in 6.5.

4.5.11 Depth to adapters or shims. This measurement shall be performed using automatic or semi-automatic inspection equipment. The design of this equipment shall be "fail-safe" to the extent that the projectile cannot be released for further processing until the proper number of shims have been inserted. Verification that the proper number of shims have been inserted shall be accomplished by re-inspection of this depth.

4.5.12 Pull test of empty plastic expulsion charge assembly (destructive test). After each sample of the lid (dwg. 9362552) and the empty cup (dwg. 9362553) have been assembled together for this inspection, the assembly shall then be subjected to this test by placing it in a suitable fixture which will restrain the cup in such a manner as to avoid any squeezing on the body of the cup. The load, in an axial direction, shall then be applied to the tab of lid, and the force required to separate the lid from the cup shall be measured. If any assembly fails to meet the minimum and maximum force requirements on the applicable drawing (9362556) the production quantity, both cup and lid, represented by that sample shall be rejected. All equipment used to perform this test shall be in accordance with 4.4.4.

5. PACKAGING

5.1 Packing

5.1.1 Level A. The projectiles shall be palleted in accordance with Dwg. 8837839.

5.2 Marking. Marking shall be in accordance with Dwg. 8837839 and Code of Federal Regulations, Title 49, Parts 100-199.

5.3 Shipping. When components from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. Not applicable to this specification.

6.2 Ordering data. See MIL-A-48078.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, USA AMCCOM ATTN: AMSMC-QAR-I(D), Dover, NJ 07801-5001.

6.4 Submission of ballistic test data and data cards. In addition to the normal distribution of records, when the projectile is procured by the U.S. Army Armament Materiel and Readiness Command, one (1) copy of all ballistic data and ammunition data cards shall be forwarded to: Commander, USA AMCCOM, ATTN: AMSMC-QAR-Q(D) Dover, NJ 07801-5001.

6.5 Data requirements. Data item descriptions required for submission under this specification are as follows:

Paragraph	Data requirement	Applicable D.I.
4.5.2, 4.5.3, 4.5.4 4.5.5, 4.5.6, 4.5.7 4.5.8, 4.5.2.1**, 4.4.2.9**, 4.4.2.10**, 4.4.2.11**, 4.4.2.14**	Quality Inspection Defect Report	DI-R-1721
4.5.1, 4.5.12	Quality Inspection Test Demonstration and Evaluation Report	DI-R-1724
6.11	Procedure Report	*DI-T-5372 (Tailored)

*Delete a, b, c, d of paragraph 1 block 10, DD Form 1664, and substitute contractor designated parameters and limits.

** Applies to critical defects only.

The above data shall be forwarded in accordance with the requirements of the contract to Commander, USA AMCCOM, ATTN: AMSMC-QAR-Q(D), Dover, NJ 07801-5001.

6.6 Projectile functioning.

6.6.1 Premature burst. A premature burst is considered to have occurred if the projectile functions in the weapon or if the time of flight is less than 1.9 seconds.

6.6.2 Functioning at all zones, excluding safety phase. Satisfactory functioning of the projectile when fired at any zone, except the safety phase, is defined as successful delivery of the cargo to the target area with all projectile hardware remaining intact until initiation of the time fuze at a pre-determined time after firing, and shall also include the successful expulsion of the entire cargo of grenades.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

6.6.3 Functioning at the safety phase. Satisfactory functioning of the projectile when fired at the safety phase is defined as successful delivery of the cargo to the target area with all projectile hardware remaining intact until initiation by the time fuze.

6.7 Grenade functioning.

6.7.1 Unsatisfactory functioning. Grenades failing to function satisfactorily are defined as those grenades which upon ground impact do not function high order, except those grenades evaluated to have been damaged by air collision/detonation.

6.7.2 Air detonation. A grenade air detonation is defined as functioning of the grenade in the air due to any cause during the time interval after expulsion from the projectile and prior to impact on the prepared field.

6.8 Barricades. Approved barricades shall be utilized for all inspections performed, when applicable.

6.9 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Armament R&D Cmd. or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.10 Automatic inspection equipment. All automatic test and inspection equipment when utilized by the contractor shall be subjected to a periodic verification or calibration check at frequent intervals during normal production. The procedures shall include as a minimum: frequency, method of verification, and the method of retrieval of items produced between verification/ calibration. These procedures shall be included in the operation/calibration procedures as required by 4.4.4.

6.11 Alternate inspection procedures.

6.11.1 Justification for alternate inspection procedures. Submitted documentation must include sufficient information to indicate an economic benefit, and must also clearly identify that there is no reduction in product quality.

6.11.2 Alternate inspection procedures. The alternate inspection procedures as a minimum must include:

a. Frequency (production interval) for validating equipment functioning.
b. Proposed method of validation to include process and inspection drawings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

c. Description of material control procedures to be imposed during the production interval between validations (Note: Under these provisions final acceptance will be withheld from material produced during the production interval until conducting satisfactory validation.)

6.11.3 Approval of alternate inspection procedures. Proposed plans shall be submitted per 6.4 for approval.

6.12 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.13 Subject term (keyword) listing.

Ammunition
Artillery
Grenades
Military Specificaiton
Munition
Projectile, 155MMM, HE: M483

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1320-A928)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-48749D (AR)	PROJRCTILE, 155MM, HE, M483A1LOADING, ASSEMBLING AND PACKING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.